INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

GRC205 Incident Management and Business Continuity, Week 21

Business Continuity and Disaster Recovery Assignment

Northbridge Skills Institute: Main Site Flood and Power Failure

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All organisations, people, roles, contact details and figures in this report are fictional exercise data. Results from the drill timings are modelled baseline results, not a live test.

Introduction

This report presents a coordinated Business Continuity and Disaster Recovery (BC/DR) plan for Northbridge Skills Institute after flooding and power failure close its main site at 08:00 WAT on Day 0, a Monday. It distinguishes temporary business services from technical restoration, uses the case data to assess priorities and costs, and compares the proposed controls with modelled recovery timings. The financial and operational assumptions are labelled A1 onwards, and proposals are marked P1 onwards. Results derived from the published drill durations are modelled baseline results, not findings from a live test. [1]

Task 1 Business Impact Analysis

1.1 Five-function BIA

The BIA follows six phases: identify critical functions, determine impacts, establish RTO and RPO, identify resources and prioritise recovery. Table 1 records each function. All five functions share the same people (90 staff and two technicians), premises (the closed main site and an alternate facility with 25 workstations from 10:00), suppliers (Microsoft 365, the payment provider, the bank, the offsite backup provider and the alternate facility provider) and information (DB01 records, AUTH01 accounts and FS01 files). Owner roles are fictional (A2).

These shared needs drive the minimum services. People are the binding constraint, because the same two technicians restore every system while business coordinators run the manual processes. Premises matter mainly for F1, since only the examination needs seats in a controlled room. Suppliers become critical during the outage: the payment provider and the bank hold the independent records that make later reconciliation possible. Information needs differ by function. F2 and F4 need complete and accurate records more than speed, whereas F1 and F3 need timely access before their deadlines.

Table 1. Business Impact Analysis for F1 to F5 (owner roles fictional, A2)

Function	Owner role	Users affected	Critical resources (people, premises, suppliers, information)	Dependencies	Impact over time	Minimum acceptable service, with approval and record controls	Priority
F1 Teaching and assessment (LMS01, DB01)	Head of Assessment	1,800 learners; 120 exam candidates (40 need a centre workstation, 80 can use own devices); tutors	Tutors, invigilators, learner support; alternate facility workstations; Microsoft 365; LMS01 data, question bank, adjustments list	Recovery network, AUTH01, DB01, LMS01; staff; secure access; communication	0-3 h: exam preparation disrupted. 11:00: exam cannot start, NGN 120,000 rescheduling and learner distress. Beyond 24 h: teaching and coursework deadlines slip.	Exam postponed by a recorded decision and re-run on validated LMS01 with an approved paper variant, invigilators and signed attendance registers; teaching notices through Microsoft 365.	1
F2 Fees and finance (FIN01, DB01)	Bursar	Fee-paying learners; finance staff	Two finance officers; payment provider portal; bank; FIN01 receipts and wallet records	Recovery network, AUTH01, DB01, FIN01; payment provider	Fee confirmation due 16:00 Day 0. Unconfirmed learners and disputes build each hour; the reconciliation backlog grows.	Payments confirmed from the provider portal; provisional confirmations from the finance mailbox; numbered manual register with two-person approval; no refunds or wallet changes until FIN01 is reconciled.	2
F3 Admissions and registration (REG01, DB01)	Registrar	Applicants; learners changing registration; admissions staff	Two admissions officers; Microsoft 365 forms and mailbox; REG01 records	Recovery network, AUTH01, DB01, REG01	Admissions close 18:00 Day 0. Applications cannot be entered; applicants may be lost and reputation harmed.	Timestamped online intake with automatic reference numbers; intake log; later entry into REG01 with a second-person check.	3
F4 Staff payroll (HR01, DB01)	HR Manager	90 staff	Payroll officer; Bursar as approver; bank portal; HR01 records	Recovery network, AUTH01, DB01, HR01; bank	Low on Day 0. Bank deadline 12:00 Day 1. Incorrect or duplicate instructions harm staff and need manual correction.	Payroll file built only from accepted HR01 data, compared with the last approved run, approved by two people and submitted once.	4
F5 Records and certificates (FS01)	Records Manager	Learners, graduates and employers requesting records	Records officer; signing authority; FS01 files	Recovery network, AUTH01, FS01	Low on Day 0. The 48-hour commitment is breached after 08:00 Day 2. Integrity or access failures are severe at any time.	Requests logged in a restricted mailbox and acknowledged; documents issued only from integrity-checked FS01 files with Registrar approval.	5

1.2 Worked impact calculations

Each function's cost is its hourly rate multiplied by the hours of disruption, applied for the first 24 hours only, with no recovery or continuity measures. The examination trigger applies in both windows because 11:00 is three hours after the disruption, so the NGN 120,000 is added once to F1. Table 2 shows the working. The totals are NGN 1,200,000 for 8 hours and NGN 3,360,000 for 24 hours. These are exercise estimates, not observed losses.

Table 2. Disruption cost without recovery (rate x hours + one-off trigger)

Function	Rate NGN/h	8-hour working	8-hour cost NGN	24-hour working	24-hour cost NGN
F1	60,000	60,000 x 8 = 480,000; + 120,000 trigger	600,000	60,000 x 24 = 1,440,000; + 120,000 trigger	1,560,000
F2	40,000	40,000 x 8	320,000	40,000 x 24	960,000
F3	20,000	20,000 x 8	160,000	20,000 x 24	480,000
F4	10,000	10,000 x 8	80,000	10,000 x 24	240,000
F5	5,000	5,000 x 8	40,000	5,000 x 24	120,000
Combined	135,000	135,000 x 8 = 1,080,000; + 120,000	1,200,000	135,000 x 24 = 3,240,000; + 120,000	3,360,000

The estimates have two main limitations. First, a flat hourly rate ignores timing. The real F2 cost is concentrated around the 16:00 deadline and the F4 cost around 12:00 on Day 1, not spread evenly. Second, the rates are treated as separate and stop at 24 hours, although the outage lasts 72 hours, and they leave out the cost of reconciliation after recovery and the shared effect of losing DB01.

The deadline effects require a decision before the technical recovery result is available. At 09:00, the examination owner should confirm the number of accessible seats, invigilator availability, integrity of examination material and a reliable contact channel for all 120 candidates. A technically restored server does not address the 15-seat deficit or guarantee equal conditions across different sittings. This is why assessment continuity is prioritised above attempting an unsafe rush to reopen LMS01.

The naira figures measure financial impact only. Learner welfare (examination anxiety, travel and accessibility), reputation (lost applicants and trust) and data integrity (wrong payroll, missing receipts) are separate impacts. They are judged qualitatively in Table 1 and can outweigh the cost figures.

1.3 Priorities and dependency risks

The business priority follows the deadlines and rates: F1 (11:00 exam, highest rate), F2 (16:00), F3 (18:00), F4 (12:00 Day 1) and F5 (48 hours). The technical restoration order is different, because F1 to F4 all need the recovery network, AUTH01, the security checks and DB01 first. Task 5 shows that the 225-minute chain to LMS01 cannot finish before 11:45. No restoration can therefore save the 11:00 exam, so F1's top priority leads to an early continuity decision rather than a faster restore. FS01 is restored early because the second technician is otherwise idle, not because F5 is urgent. Table 3 lists the dependency risks.

Table 3. Important dependency risks

ID	Dependency risk	Effect	Treatment
D1	DB01 is the single data base for F1 to F4	One corrupt or failed restore stops four functions	Verified restore chain, 30-minute incrementals and monthly restore tests (Table 6)
D2	AUTH01 snapshot is 8 hours old	Accounts, passwords and permission changes made after 00:00 are lost, which affects access to every function	Two-hourly snapshots; re-apply logged changes after restoration
D3	Production and backup administration share one identity	One compromised account can damage both production and backups	Separate identities with MFA; break-glass account (CA1)
D4	The offsite provider's access-independence controls are undocumented	Backups may be reachable by the same attacker or unavailable when needed	Written attestation and a witnessed test (CA4)
D5	Only two technicians	Sickness or fatigue delays every restore	Cross-train a third person (CA8)
D6	25 workstations for 40 centre learners	The exam cannot run in one sitting	Split sittings (Table 14); surge contract (CA5)

Task 2 Recovery objectives and backup design

2.1 Definitions and targets

The Recovery Time Objective (RTO) is the maximum time a function or system can remain unavailable before the impact becomes unacceptable. The Recovery Point Objective (RPO) is the maximum acceptable data loss, measured in time before the disruption. The Maximum Tolerable Downtime (MTD) is the total outage the owner is willing to accept, taking all impacts into account. NIST states that the RTO must normally be shorter than the MTD, because reprocessing time is added after recovery. The usual definition measures RTO up to full restoration. In this report, recovery is declared when the minimum acceptable service in Table 1 is accepted by the function owner, as the brief requires (P1). Full restoration then follows through the controlled return described in Section 4.6.

Table 4 sets the targets (P2). Each MTD comes from a stated deadline. Each RTO leaves a buffer for business acceptance, reconciliation and delay. The F2 buffer is the smallest, at three hours.

Table 4. Recovery objective matrix (targets are proposals, P2)

Function	RTO	RPO	MTD	Buffer (MTD - RTO)	Justification
F1	5 h	1 h	24 h	19 h	The 11:00 exam (3 h) is handled by rescheduling, which no RTO can meet. LMS01 is needed for teaching and for the Day 1 sitting. The 5 h RTO matches the upper warm estimate.
F2	5 h	1 h	8 h	3 h	The 16:00 confirmation deadline is 8 h after the disruption. The buffer covers acceptance and reconciliation before 16:00.
F3	6 h	1 h	10 h	4 h	Admissions close at 18:00, 10 h after the disruption. The buffer allows online-intake records to be keyed in before closing.
F4	12 h	4 h	24 h	12 h	The bank deadline is 28 h away; 4 h is reserved for checks and two-person approval, giving an MTD of 24 h. Payroll data changes slowly.
F5	24 h	4 h	48 h	24 h	The 48-hour service commitment sets the MTD; file integrity matters more than speed.
AUTH01 (support)	Before DB01	4 h	As F1-F3	-	A prerequisite for every function, so it is restored first.

2.2 Checkpoint analysis

Table 5 measures each checkpoint's age from the 08:00 disruption. DB01 meets the 1-hour RPO for F1 to F3, but with no margin. AUTH01 exceeds its 4-hour RPO by 4 hours, and FS01 exceeds its 4-hour RPO by 5 hours.

The DB01 and AUTH01 checkpoints must be assessed separately. Although F1–F4 depend on a one-hour-old database point, the directory restored from 00:00 may lack eight hours of permission, password and account changes. Recreating those changes without evidence could grant inappropriate access. The ICT Manager should compare administrator audit records, rebuild only approved changes, and obtain independent sign-off before the restored directory is trusted for learner or finance access.

Table 5. Age of the latest recoverable checkpoints at 08:00 Day 0

Item	Checkpoint	Working	Age	Functions	RPO	Gap
B01 + B02 (DB01)	07:00 Day 0	08:00 - 07:00	1 h	F1, F2, F3, F4	1 h (F1-F3); 4 h (F4)	DB01 data checkpoint meets F1–F3 1 h and F4 4 h targets; supporting AUTH01 gap remains
B03 AUTH01	00:00 Day 0	08:00 - 00:00	8 h	All (access)	4 h	4 h over
B04 FS01	23:00 Day -1	24:00 - 23:00 + 8 h	9 h	F5	4 h	5 h over
Disconnected full	17:00 Day -3	7 h + 48 h + 8 h	63 h	Fallback (A10)	-	Last resort only

Some data can be rebuilt from independent records. Payments can be reconciled from the payment provider's export. Applications and requests made after 08:00 are held in the temporary intake logs, and payroll can be checked against the last approved run and the approval emails. Other data cannot be assumed recoverable: DB01 transactions entered between 07:00 and 08:00 (receipts, wallet changes, registration changes and LMS submissions), AUTH01 changes made after 00:00, and FS01 files created after 23:00 on Day -1. Owners must ask users to re-submit these items and record each exception.

2.3 Backup and restoration design

Table 6 proposes a protected 3-2-1 arrangement: three recoverable copies on at least two storage-media types, with one copy offsite and a separately controlled immutable or offline copy (P3). To restore DB01, begin with the most recent verified full, apply any selected differential based on that full, and then apply every subsequent required incremental in order. Under the proposed Monday 08:00 design, this means the Sunday 23:00 full plus 17 half-hour incrementals from 23:30 to 07:30. The resulting checkpoint is 30 minutes old, inside the one-hour proposed RPO; this is not the existing B01/B02 chain, which reaches only 07:00. [2, slide 8]; [3, sec. 3.4]

Table 6. Backup schedule and protection design (proposals, P3)

System	Full	Incremental or differential	Retention	Placement (3-2-1)	Protection	Restore test
DB01	Weekly, Sunday 23:00	Differential daily at 23:00 (Mon-Sat); incremental every 30 min	Incrementals 14 days; differentials 35 days; weekly fulls 13 weeks; month-end fulls 12 months	Three copies across production disk, independent encrypted offsite disk, and separately administered immutable cloud-object or offline removable media at another location (at least two media types)	Encryption in transit and at rest; decryption keys separately held; dedicated backup identities with MFA; separate production access; immutability protected from backup operators	Monthly restore into an isolated environment
AUTH01	Weekly system-state backup	Snapshot every 2 h and before any administrative change	30 days; month-end 12 months	As DB01	As DB01; break-glass account sealed	Quarterly
FS01	Weekly, Sunday 23:00	Incremental every 2 h	Incrementals 30 days; fulls 13 weeks; month-end 12 months	As DB01	As DB01; access lists preserved	Quarterly, plus file hash sampling

Neither offsite storage nor a live replica proves recoverability on its own. An offsite copy can be incomplete, corrupt, encrypted with unavailable keys, or reachable through the same compromised identity. A replica copies corruption and malicious deletions almost immediately. NIST advises testing backups regularly so that files can be retrieved without errors or lost data. Only a successful restore followed by application and data validation proves that a backup can be recovered. Table 7 defines the checks. The baseline assumes that B01 to B04 pass their integrity checks, as the brief states. The provider's access-independence remains unverified (A6, CA4).

A successful integrity check is only one condition for restoration. The organisation must also demonstrate that the backup can be accessed without a compromised production account, that encryption keys remain available, that the chain contains every required increment, and that data can be opened by restored applications. A witnessed test should retain the source checkpoint, last transaction time, selected restoration chain, exception list and function-owner sign-off. This provides evidence of usable recovery rather than relying on a supplier label or a successful backup-job notification.

Table 7. Verification of backup integrity, access independence and usability

Check	Method	Evidence	Frequency
Integrity	Checksum each backup job; confirm the catalogue shows an unbroken chain to the target point	Job report with verified hashes; signed chain list	Every job; before any restore
Access independence	Separate backup identities with MFA; provider attestation; review of backup-console logons	Access review; signed attestation; logon report	Monthly; quarterly attestation
Usability	Restore to an isolated environment; database consistency check; test login; sample records compared with source reports; last transaction time	Signed restore test record	Monthly (DB01); quarterly (AUTH01, FS01)

Task 3 Recovery strategy and budget

3.1 Option comparison

Table 8 compares the packages against the targets in Table 4. The quoted windows are supplier planning estimates, not guarantees. The general site characteristics follow the standard descriptions of cold, warm and hot sites.

Geography, staffing and supplier dependency separate the options as much as speed does. None of the packages states where its recovery site is, so all three carry the same unconfirmed risk that the site sits in the same flood area or on the same power supply (A6). Cold would ask two technicians to build and configure systems under time pressure. Hot moves the work to continuous replication, which the provider must run correctly every day. Warm restores from backups that Northbridge controls, so a verified restore point can be chosen before data is loaded. This matters for propagation risk, because replication copies corruption or malicious changes as quickly as it copies good data.

Table 8. Option matrix (supplier figures are estimates)

Criterion	Cold	Warm	Hot
Estimated first critical service	24-48 h	3-5 h	About 1 h
Against RTOs (5 h F1-F2, 6 h F3)	Fails F1-F3 and the F2 and F3 MTDs	Meets them at the upper estimate only	Meets them, and may meet the 11:00 exam if validated
Data-loss window against 1 h RPO	Up to 24 h: fails	1-6 h: fails unless Northbridge's own 30-minute chain is used	Up to 15 min: meets
Staffing (two technicians)	Heavy build effort; slowest	Feasible: modelled drill completes in 4 h 15 min	Lower recovery effort, but continuous monitoring is needed
Geography	Location unknown (A6)	Location unknown (A6)	Location unknown (A6)
Supplier dependency	Low during normal operation; high at recovery	Moderate	High and continuous
Propagation of corruption or malicious change	Low (offline restore)	Moderate (restore from a verified point)	High (near-real-time replication copies changes)

3.2 Budget

Table 9. Itemised annual cost of each package (NGN)

Package	Base	Independent protected backups	Training and tests	Total	Limit 3,000,000 minus total	Within limit
Cold	600,000	400,000	200,000	1,200,000	1,800,000	Yes
Warm	1,500,000	400,000	200,000	2,100,000	900,000	Yes
Hot	2,400,000	400,000	200,000	3,000,000	0	Yes, exactly at the limit with no contingency

3.3 Recommendation, residual risk and investment reasoning

Warm is the preferred package at NGN 2,100,000 annually, leaving NGN 900,000, or 30% of the NGN 3,000,000 planning limit, uncommitted. Cold is too slow for the immediate finance and admissions deadlines. Hot reaches the entire budget ceiling and requires independent testing of the fast recovery and replication claims. Warm also cannot support the original 11:00 examination, and its quoted 1–6-hour data-loss window cannot independently meet the proposed one-hour RPO for F1–F3. The examined backup schedule and controlled exam postponement are therefore necessary. Table 10 records risks to be accepted by the Principal after consultation with the function owners and ICT Manager. [1, pp. 3–5]

Table 10. Residual risks and acceptance

Residual risk	Treatment	Accepted by
Rescheduling cost (NGN 120,000) whenever an outage falls on an exam morning	Exam contingency procedure (CA6)	Principal, on advice of the Head of Assessment
RPO depends on Northbridge's own backup chain, not the package	30-minute incrementals with integrity checks (CA3)	Principal, on advice of the ICT Manager
Supplier timings are unvalidated	Contract test clause before payment	Principal and Bursar
Two-technician dependency	Cross-training (CA8)	Principal
Workstation shortfall (25 for 40 learners)	Surge contract (CA5)	Principal

Warm costs NGN 2,100,000 per year compared with NGN 1,200,000 for eight hours or NGN 3,360,000 for 24 hours of uninterrupted disruption without workarounds. These are different types of estimates: the strategy price is an annual recurring cost, while the disruption figures describe hypothetical single-event losses. Neither the comparison nor the NGN 120,000 examination charge establishes an annual return on investment. Decision makers need credible estimates of outage frequency, avoided loss, supplier exclusions and operational benefits, as well as a witnessed warm-site restoration, verified site geography, contracted service levels, access-independence evidence and competing quotations. [1]

Task 4 Business continuity and disaster recovery plan

4.1 Activation, roles and coordination

Table 11 sets out the core elements of the plan, following the standard plan structure and NIST's activation criteria. The plan works under a unified command led by the Principal. The continuity team (the BC/DR Coordinator and the function owners) runs the temporary procedures and business acceptance. The technical recovery team (the ICT Manager and two technicians) follows the runbook. Facilities manages the alternate facility and site clearance. The incident response team stays on standby. All teams report through one situation report, and an IR request to pause restoration for evidence preservation is decided and recorded by the Principal. The BC/DR Coordinator keeps the decision log and chairs a short coordination call before each situation report, so that technical progress, temporary procedures and communications stay consistent.

Table 11. Plan scope, activation and reporting

Element	Content
Scope	F1-F5, their systems, the main site, the alternate facility and the warm recovery site; all-hazards disruption, with this 72-hour outage as the reference case
Activation thresholds (P4)	Main site, DB01 or AUTH01 unavailable, or expected to be unavailable, for more than 2 hours; an exam within 24 hours under threat; any F2-F4 deadline at risk; any compromise indicator (also activates IR)
Decision maker and deputy	Principal; Deputy Principal as deputy (fictional roles)
Escalation	Coordinator to Principal within 15 minutes of a threshold being met; Principal to Steering Committee if an MTD is at risk or unbudgeted spend exceeds NGN 250,000 (P4); ICT Manager to IR Lead on any compromise indicator
Contact roles	Principal, Deputy Principal, BC/DR Coordinator, ICT Manager, Technicians 1 and 2, function owners, Facilities Manager, Communications Lead, IR Lead, Data Protection Officer, supplier account managers; mailbox bcdr@northbridge.example (fictional)
Alternative communication	Primary: Microsoft 365 email and Teams over mobile internet. Fallback: SMS and voice call tree from staff mobile phones (A13); printed contact cards held by role holders
Situation reporting	09:00, 12:00 and 16:00 on Day 0, then 09:00 and 16:00 daily. Each report gives function status against RTO and MTD, decisions, risks, next actions and communications issued

4.2 RACI matrix

Table 12 assigns a single accountable role and at least one responsible role to each activity. The BC/DR Coordinator is accountable for the overall service-acceptance process; individual function owners must still sign the acceptance evidence before their services resume. [2, slide 12]

Table 12. RACI matrix. PR Principal; BC BC/DR Coordinator; ICT ICT Manager; TEC Technicians; HA Head of Assessment; FO Owners of F2-F5; FM Facilities Manager; CL Communications Lead; IR IR Lead; DPO Data Protection Officer. A/R = accountable and responsible

Activity	PR	BC	ICT	TEC	HA	FO	FM	CL	IR	DPO
1 Plan activation	A	R	C	I	C	I	C	I	I	I
2 Temporary assessment delivery	I	C	C	I	A/R	-	R	C	-	C
3 Backup selection and validation	I	I	A	R	-	-	-	-	C	-
4 System restoration	I	I	A	R	I	I	-	-	C	-
5 Business acceptance of restored service	I	A	C	C	R	R	-	-	-	-
6 Stakeholder communication	A	C	C	-	C	C	-	R	C	C
7 Compromise assessment and IR escalation	I	I	R	C	-	-	-	I	A/R	C
8 Return to primary site and final sign-off	A	R	R	C	C	C	R	I	C	-

4.3 Temporary procedures for the 72-hour outage

At a 09:00 go/no-go point (P9), the Principal, advised by the Head of Assessment, postpones the 11:00 examination. The earliest modelled LMS01 restoration is 11:45, before business acceptance, and only 25 workstations exist for the 40 learners who need one. Starting a partly validated exam would put integrity and accessibility at risk, so the NGN 120,000 trigger is accepted (A4). The exam is re-run on Day 1 in two sittings with different approved paper variants (Table 14; A3, A5). Learners with adjustments are seated in the first sitting, with assistive software checked on Day 0. Workstation profiles are wiped between sittings, and business staff work on laptops through Microsoft 365 so that the workstations stay free for learners (A14). Table 13 sets out the procedure for each function.

Table 13. Temporary procedures F1 to F5 (staffing numbers are proposals, P5; staff availability A11)

Function	Temporary procedure	Approvals and record controls	Integrity, privacy and accessibility	Reconciliation	Minimum staff, equipment, suppliers
F1	Postpone the exam; teach through Microsoft 365 notices; re-run on Day 1 (Table 14)	Principal approves postponement; Head of Assessment and moderator approve variant B; signed attendance and incident registers	Separate variants; invigilation; no personal storage; wiped profiles; adjustments list honoured	Registers matched to LMS01 submissions; incident log reviewed	Head of Assessment; 4 invigilators or proctors; 2 learner support staff; 25 workstations; LMS01; facility supplier
F2	Confirm payments in the provider portal; send provisional confirmations; numbered manual register in a restricted Microsoft 365 list; no refunds or wallet changes	Finance officer prepares, Bursar approves; entries timestamped	Register restricted to finance; payer data minimised; MFA	Every provider transaction since 07:00 matched to FIN01; missing receipts posted as approved adjustments; exceptions log	Bursar; 2 finance officers; 2 laptops; provider portal
F3	Online intake form with automatic reference; registration changes through the admissions mailbox; 18:00 closing judged by timestamp	Intake log; Registrar decides late or unclear cases	Restricted form; secure document upload; telephone support for accessibility	Entries keyed into REG01 with a second-person check; log totals equal REG01 totals	Registrar; 2 admissions officers; Microsoft 365
F4	No payroll action on Day 0. If HR01 is not accepted by 08:00 Day 1, submit the last approved payroll plus approved variations	HR Manager and Bursar approve; fallback also needs Principal approval; one submission reference	Payroll data restricted; secure bank portal	HR01 compared with the last approved run and logged changes; duplicate check before submission	HR Manager; payroll officer; Bursar; bank portal
F5	Log requests in a restricted mailbox and acknowledge within 4 working hours (P5)	Registrar approves each issue; issue register	Requester identity checked; file hashes compared before issue	Files changed after 23:00 Day -1 re-checked; lost uploads re-requested	Records Manager; 1 records officer

Table 14. Baseline Day 1 examination seating (proposal P9; A3, A5, A7)

Sitting	Time (Day 1)	Centre learners	Workstations used	Own-device learners	Paper variant	Invigilation
1	09:00	25	25 of 25	80 (remote, A7)	A	2 room invigilators + 2 remote proctors
2	14:00	15	15 of 25 (10 free for staff)	0	B	2 room invigilators
Total	-	40	-	80	-	120 candidates seated

4.4 DR runbook

Steps 1 to 11 in Table 15 restore technology. Steps 12 and 13 resume an acceptable business service. A restored LMS01 is not an assessment service until the Head of Assessment accepts it.

Table 15. DR runbook

Step	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
1	Confirm activation, assign technicians, open recovery log	Activation decision	ICT Manager	Log open; roles confirmed	Escalate to BC/DR Coordinator
2	Select backups: B01 + B02 chain to 07:00, B03, B04; record hashes	1	ICT Manager	Catalogue shows an unbroken chain; selection signed	Use the latest verified point; if tampering is suspected, call IR and consider the Day -3 disconnected full (A10)
3	Recovery network (30 min)	None	Technician 1	Links to backup source and alternate facility work	Supplier escalation; alternate link
4	AUTH01 restoration (30 min)	3	Technician 1	Directory responds; shared admin identity disabled; new admin credentials issued	Rebuild essential accounts from documented configuration; IR if anomalies appear
5	Security and backup checks (30 min)	4	Technician 1	Restored AUTH01 scanned clean; backup-console logons reviewed; B02 checksums verified	Halt the affected restore; call IR; choose an earlier verified point
6	DB01 restoration: B01, then B02 in order to 07:00 (90 min)	5	Technician 1	Consistency check passes; last transaction about 07:00; record counts match the last reports	Restore to the last verified incremental; roll back to the pre-apply snapshot
7	FS01 restoration (60 min)	4 and 5	Technician 2	Hash sample matches manifest; access lists unchanged	F5 stays manual; try an alternative copy
8	LMS01 restoration (45 min)	6	Technician 1	Login, course page and test submission on a test account	Roll back the application snapshot; Day 1 exam plan unchanged
9	FIN01 restoration (30 min)	6	Technician 2	Test receipt on a test ledger	Continue the manual register
10	HR01 restoration (45 min)	6	Technician 2	Staff count equals the last payroll; no pending batch	F4 fallback
11	REG01 restoration (30 min)	6	Technician 1	Test application record created and removed	Continue online intake
12	Business acceptance of each function	7-11	Function owner	Signed checklist: sample records, last transaction time, access rights; minimum service declared	Reject; roll back; remain on the manual procedure
13	Reconcile the gap and temporary records; full backup of the restored state	12	Owners; technicians	Exceptions closed or escalated; backup verified	IR if differences are unexplained

4.5 Communication

Table 16 sets out the communication matrix. The initial learner notice follows it.

Table 16. Stakeholder communication matrix

Stakeholder	Information need	Channel (primary / alternative)	Owner	Timing
120 exam candidates	Postponement, new arrangements, adjustments	Email / SMS (A13)	Head of Assessment via Comms Lead	By 09:30 Day 0; arrangements by 18:00
All learners	Service status; how to pay, apply and request records	Email / SMS	Comms Lead	10:00 Day 0, then daily
Staff	Roles, where to report, remote working	Teams / call tree	BC/DR Coordinator	08:30 Day 0 and at each situation report
Applicants	How to apply before 18:00	Email / SMS	Registrar	10:00 Day 0
Payment provider	Transaction export; reconciliation support	Email / telephone	Bursar	10:00 Day 0 and after FIN01 acceptance
Bank	Payroll timing or any delay	Relationship manager	HR Manager	By 16:00 Day 0 if at risk
Alternate facility and recovery providers	Activation, capacity, extension	Telephone / email	Facilities Manager; ICT Manager	08:15 Day 0
Executive Steering Committee	Situation reports, decisions, risk acceptance	Teams / email	Principal	Each situation report
Data protection regulator	Notification only if a notifiable personal data breach occurs	Official channel	DPO with Principal approval	Within 72 hours of awareness
Media and public	Holding statement	Approved statement	Comms Lead with Principal approval	On enquiry

> **Subject: Main site closed today - 11:00 examination postponed**
>
> Dear learner,
>
> Flooding and a power failure closed the Northbridge main site at 08:00 today, Monday. Please do not travel to the main site.
>
> Today's 11:00 examination will not take place, and you will not be penalised for this change. By 18:00 today we will send you the new date, time and venue, together with the arrangements for learners using their own devices.
>
> If you need a centre workstation, extra time or other support, please reply to this message or email learnersupport@northbridge.example by 14:00 today so that we can arrange your place.
>
> Fee payments made through our payment provider remain valid. We will send an online form for admissions and registration requests by 10:00.
>
> We will update you by email and text message each day until normal services return. We are sorry for the disruption and thank you for your patience while we restore our services.
>
> Learner Support Team, Northbridge Skills Institute (fictional)

4.6 Incident response trigger and controlled return

No cyberattack has been confirmed. The IR Lead is activated, while recovery continues under the IR Lead's direction for the affected components, if any of the following occurs: an unexpected administrator or backup-console logon, a backup integrity failure with no benign explanation, data changes or missing transactions that the checkpoint times do not explain, malware alerts, or a ransom demand (A12). The DPO then assesses whether a personal data breach is likely to put individuals' rights at risk. If it is, the regulator must be notified within 72 hours of awareness, and high-risk breaches must be communicated to the people affected (A1).

The return to the primary site follows NIST's reconstitution approach of validating data and functions before the plan is deactivated. Table 17 defines the controlled transfer of new records and final sign-off.

Table 17. Return to the primary site and final sign-off

Step	Criterion and evidence	Owner
1 Facilities clearance	Structural, electrical and flood-damage clearance certified in writing	Facilities Manager
2 Primary infrastructure ready	Power, network and servers tested; restored images verified	ICT Manager
3 Transaction freeze	Announced cut-off time; alternate systems set to read-only	BC/DR Coordinator
4 Transfer of new records	Final verified backup at the alternate site; restore to primary; record counts and checksums match	ICT Manager
5 Business validation	Each owner signs an acceptance checklist; temporary records fully reconciled	Function owners
6 Final sign-off and deactivation	Principal signs after steps 1-5; users notified; alternate copy kept read-only for 7 days (P6); review within 10 working days	Principal

Task 5 Recovery schedule and exercise

5.1 Baseline recovery schedule (modelled)

Table 18 uses the prescribed warm-site activity durations and two technicians. No activity starts before its prerequisites are complete, no technician has overlapping assignments and no action is split. The earliest technical LMS01 finish is 11:45, 45 minutes after the planned examination. Its critical path is the recovery network (30 minutes), AUTH01 (30), security and backup checks (30), DB01 (90) and LMS01 (45): 225 minutes after 08:00. All five technical services can finish by 12:15. [1, p. 4]

Table 18. Baseline recovery schedule (modelled, not a live test)

Action	Technician	Start	Finish	Minutes	Prerequisite
Recovery network	T1	08:00	08:30	30	None
AUTH01 restoration	T1	08:30	09:00	30	Recovery network
Security and backup checks	T1	09:00	09:30	30	AUTH01 restoration
DB01 restoration	T1	09:30	11:00	90	Security and backup checks
FS01 restoration	T2	09:30	10:30	60	AUTH01 restoration and security and backup checks
LMS01 restoration	T1	11:00	11:45	45	DB01 restoration
FIN01 restoration	T2	11:00	11:30	30	DB01 restoration
HR01 restoration	T2	11:30	12:15	45	DB01 restoration
REG01 restoration	T1	11:45	12:15	30	DB01 restoration

The earliest possible time for completing all five technical functions is 12:15 under these fixed durations. DB01 cannot finish before 11:00 because the common prerequisite chain takes 180 minutes. The remaining LMS01, FIN01, HR01 and REG01 actions total 150 technician-minutes; with two technicians, these require at least another 75 minutes. The allocation in Table 18 reaches that lower bound by assigning 45 + 30 minutes to each technician after DB01 completes. FS01 can run separately from 09:30 to 10:30 on the second technician. This is a schedule calculation, not an observed recovery result (Appendix C).

5.2 Comparison with objectives

Table 19 compares modelled technical finish times and the additional 30-minute business-owner acceptance allowance (A8) with proposed RTO and MTD thresholds. All five functions meet the proposed recovery-time limits under the baseline assumptions, although F1 cannot meet the original 11:00 examination. DB01 meets the application-data RPO for F1 to F4, but the eight-hour-old AUTH01 snapshot exceeds the four-hour supporting-directory target. F5 also misses its FS01 RPO: nine hours of potential data loss against a four-hour target. Accordingly, the baseline does not establish that every recovery-point objective is met.

Table 19. Modelled baseline results against objectives (not a live test)

Function	Technical finish	Elapsed from 08:00	Declared (+30 min, A8)	RTO / MTD	Checkpoint age / RPO	Result
F1	11:45	3 h 45 min	12:15 (4 h 15 min)	5 h / 24 h	1 h / 1 h	RTO/MTD and DB01 RPO met; AUTH01 RPO gap; exam postponed
F2	11:30	3 h 30 min	12:00 (4 h)	5 h / 8 h	1 h / 1 h	RTO/MTD and DB01 RPO met; AUTH01 RPO gap remains
F3	12:15	4 h 15 min	12:45 (4 h 45 min)	6 h / 10 h	1 h / 1 h	RTO/MTD and DB01 RPO met; AUTH01 RPO gap remains
F4	12:15	4 h 15 min	12:45 (4 h 45 min)	12 h / 24 h	1 h / 4 h	RTO/MTD and DB01 RPO met; AUTH01 RPO gap remains
F5	10:30	2 h 30 min	11:00 (3 h)	24 h / 48 h	9 h / 4 h	RTO/MTD met; FS01 RPO misses by 5 h; AUTH01 gap

Business continuity arrangements remain necessary even when the technical schedule succeeds. The 11:00 examination precedes the earliest LMS01 restoration by 45 minutes, and 25 centre workstations cannot seat all 40 centre-dependent learners at once. Fees must still be confirmed by 16:00, admissions close at 18:00, and payroll instructions are due by 12:00 on Day 1. Temporary logs and independent payment-provider records must be reconciled against the restored applications. Changes to DB01 after 07:00, AUTH01 after midnight and FS01 after 23:00 on Day minus 1 cannot be assumed recoverable. [1, pp. 2–4]

5.3 Tabletop exercise

The 45-minute tabletop is recorded as an individual role-based simulation. Minutes 0–9 establish the baseline and response objectives; the scenario changes are introduced at minutes 10, 20, 30 and 40, followed by a closing review. Each recorded decision identifies an owner, reason, revised action and effect on service. The failed-backup stress scenario is evaluated separately from the modelled baseline restoration schedule. [1, p. 7]; [2, slide 11]

Table 20. Exercise log (stress scenario, simulated roles)

Minute	Inject	Decision	Revised action	Owner	Rationale	Effect on service or recovery
10	Closure extends from 72 hours to five days	Extend the alternate facility and recovery-site booking to 7 days; move the return date	Facilities Manager confirms the extension; staff rota for 5 days; daily learner updates; F5 manual service extended past the 48-hour commitment	Facilities Manager	Recovered services must be sustained, not only restored	No change to restoration times; continuity costs rise beyond the 24-hour estimate window
20	Only 15 working workstations	Three sittings with three paper variants	Day 1 09:00 and 14:00 sittings with 15 learners each; Day 2 09:00 sitting with 10 learners; adjustment learners in sitting 1; own-device learners unchanged	Head of Assessment	Integrity is kept through variants; accessibility seats come first	The exam finishes one day later; staff lose spare workstations
30	DB01 incremental fails its integrity check; unexpected backup administrator logon	Pause the DB01 restore; activate IR; preserve logs; disable the shared identity	Preserve the backup logs and isolate suspicious administrator access. IR verifies whether a complete, trustworthy B01-to-B02 chain can be rebuilt. Use the latest verified contiguous checkpoint only if investigation clears it; otherwise evaluate B01 alone or the disconnected Day-minus-3 copy, and update the restoration schedule.	IR Lead (decision recorded by Principal)	Restoring from a possibly tampered chain could spread compromise	Checkpoint age becomes 9 h with B01 alone (8 h beyond a 1 h RPO; 5 h beyond F4’s 4 h RPO), or 63 h with the disconnected copy (62 h beyond a 1 h RPO). Incident investigation delays recovery; actual duration is unknown.
40	Payment provider reports payments missing from restored finance records	Treat the provider export as the authoritative record of payments	Match each missing payment by reference and time; post approved adjustments; learner confirmations stand; payments not explained by the checkpoint gap go to IR	Bursar	Payments in the lost window are expected; unexplained gaps may indicate tampering	FIN01 accepted only after reconciliation; affected learners notified

The exercise showed that the plan's weakest points are its dependencies, not its timings. Injects 3 and 4 showed that one doubtful backup chain or administrator identity can turn a facilities outage into a possible security incident, and that the payment provider's records are the only independent check on finance data. Inject 2 showed that seating, rather than technology, limits the examination. The corrective actions in Table 21 target these findings, and each has evidence that can be checked.

The third inject is especially significant because a failed incremental cannot automatically be bypassed while applying later dependent incrementals. A restore point is credible only if the full backup and every necessary subsequent part of its chain pass validation. The alternate checkpoint also changes the RPO outcome independently of the unknown investigative delay to the RTO. The fourth inject requires reconciliation against provider transaction identifiers before FIN01 can be accepted, with unexplained discrepancies escalated rather than silently overwritten.

Table 21. Corrective actions (proposals P10; target dates relative to fictional plan approval date P, A17)

ID	Action	Priority	Owner	Target date	Measurable completion evidence
CA1	Separate backup administration identities with MFA; remove the shared identity	High	ICT Manager	P + 30 days	Access review shows zero shared identities and 100% MFA enrolment
CA2	Add an immutable or offline third copy with a 35-day lock	High	ICT Manager	P + 60 days	Successful restore from the immutable copy; lock settings recorded
CA3	Move DB01 to 30-minute incrementals with automatic integrity checks and alerts	High	ICT Manager	P + 45 days	30 consecutive days of job reports with at least 99% verified
CA4	Obtain and test the offsite provider's access-independence controls	High	ICT Manager with Bursar	P + 60 days	Signed attestation and a witnessed test record
CA5	Contract surge capacity of at least 40 workstations, with a second facility option	Medium	Facilities Manager	P + 90 days	Signed addendum; walkthrough confirms 40 working workstations
CA6	Approve the exam contingency procedure with banked variants and a 09:00 go/no-go rule	High	Head of Assessment	P + 30 days	Approved procedure; two banked variants for every exam this term
CA7	Write a payment reconciliation procedure with daily provider exports	Medium	Bursar	P + 45 days	Test reconciliation completed with zero unexplained items
CA8	Cross-train a third recovery person	Medium	ICT Manager	P + 120 days	Supervised DB01 restore completed within the drill time

Task 6 Governance and performance measures

6.1 Ownership, testing and maintenance

Following the standard governance model, the BC/DR Coordinator owns and maintains the plan, and the Principal approves it on behalf of the Executive Steering Committee each year and after any major change. Function owners maintain their own procedures, and the ICT Manager maintains the runbook. Training is given at least annually, as NIST advises, and within 30 days for anyone newly appointed to a role (P7). Tables 22 and 23 set out the controls and the testing calendar.

Table 22. Governance controls (proposals, P7)

Control	Proposal
Version control	Version number, change log and approval record; superseded copies withdrawn
Distribution	Controlled list of role holders; Microsoft 365 copy plus an encrypted offline copy; printed quick-reference cards at the alternate facility
Review triggers	Any activation or test; system, supplier or staff-role change; flood warnings; new legal requirements; annual review at a minimum
Training	Annual training for all role holders; induction within 30 days; records held by the HR Manager

Table 23. Risk-based testing calendar (proposals, P7; frequencies follow standard practice where stated)

Activity	Timing	Focus	Owner
DB01 restore test	Monthly	Highest-impact shared data	ICT Manager
AUTH01 and FS01 restore tests	Quarterly	RPO gaps closed	ICT Manager
Tabletop exercise	Quarterly; one before each exam period	Exam and deadline decisions	BC/DR Coordinator
Walkthrough test	Semi-annually	Runbook and temporary procedures	BC/DR Coordinator
Simulation with warm-site parallel test	Annually, before the rainy season	Supplier timings and RTO evidence	ICT Manager
Call-tree test	Quarterly	Alternative communication	Comms Lead

6.2 Framework mapping and climate consideration

Table 24 maps the recovery programme to all seven NIST SP 800-34 contingency-planning steps and the operational themes of ISO 22301:2019, including leadership, planning, operation, evaluation and improvement. It demonstrates relevant design alignment, not ISO certification or an implementation audit. [2, slides 4 and 9]; [3]; [5]

Table 24. Framework mapping (educational alignment only)

Framework	Step or clause (standard wording)	Report output
NIST SP 800-34	01 Develop Contingency Planning Policy Statement	Ownership and approval (Section 6.1)
NIST SP 800-34	02 Conduct Business Impact Analysis (BIA)	Tables 1-3
NIST SP 800-34	03 Identify Preventive Controls	Backup design and access controls (Tables 6-7)
NIST SP 800-34	04 Create Contingency Strategies	Option matrix and budget (Tables 8-10)
NIST SP 800-34	05 Develop Information System Contingency Plan (ISCP)	DR runbook (Table 15)
NIST SP 800-34	06 Ensure Plan Testing, Training & Exercises (TT&E)	Schedule and tabletop (Tables 18-21)
NIST SP 800-34	07 Plan Maintenance	Governance and testing calendar (Tables 22-23)
ISO 22301	Clause 4 Context	BIA and climate consideration
ISO 22301	Clause 5 Leadership	Decision maker, deputy and risk acceptance (Tables 10-11)

Framework	Step or clause (standard wording)	Report output
ISO 22301	Clause 6 Planning	Recovery objectives (Table 4)
ISO 22301	Clause 7 Support	Training and communication (Tables 16, 22)
ISO 22301	Clause 8 Operation	Strategy, plan and temporary procedures (Tasks 3-4)
ISO 22301	Clause 9 Performance	KPI dashboard (Table 25)
ISO 22301	Clause 10 Improvement	Corrective actions (Table 21)

This disruption was caused by flooding. ISO's 2024 climate action amendment to ISO 22301 adds a requirement to clause 4.1 that the organisation determine whether climate change is a relevant issue, and a note to clause 4.2 that interested parties can have climate-related requirements. Northbridge should record climate change as relevant. In practice, the alternate facility, the warm site and the backup locations should sit outside the main site's flood area and power supply (A6), local servers should be moved above flood level, rainfall warnings should act as a pre-activation alert, and the annual simulation should run before the rainy season. NIST likewise lists the risk of the offsite storage site being hit by the same disaster as a selection criterion.

6.3 KPIs and maturity

Table 25 defines the performance dashboard. Its targets are proposals (P8). The recovery, testing and compliance categories guide the choice of measures. K1 and K2 compare actual recovery with the targets in Table 4, K3 shows whether restore tests cover every critical system, K4 tracks the closure of exercise findings and K5 measures training. K6 adds the daily backup evidence on which every restore depends.

Table 25. Performance dashboard (targets are proposals, P8)

KPI	Formula or measurement method	Data source	Target	Frequency	Accountable owner	Current baseline
K1 Recovery time performance	Functions declared within RTO / functions tested x 100	Exercise and test logs	100%	Quarterly	BC/DR Coordinator	5/5 modelled (not live)
K2 Recovery point performance	Systems whose restored checkpoint age is within RPO / systems tested x 100	Restore test records	100%	Monthly	ICT Manager	1/3 (DB01 only; AUTH01, FS01 miss)
K3 Restoration-test coverage	Critical systems successfully restored in the period / 7 critical systems x 100	Restore test register	100% each quarter	Quarterly	ICT Manager	Not yet measured
K4 Corrective-action closure	Actions closed by target date / actions due x 100	Corrective-action register	At least 90%	Monthly	BC/DR Coordinator	0 of 8 due
K5 Training completion	Role holders trained in the last 12 months / all role holders x 100	HR training records	100%	Quarterly	HR Manager	Not yet measured
K6 Backup integrity	Jobs passing integrity verification / all jobs x 100	Backup reports	At least 99%	Weekly	ICT Manager	Not yet measured

On the maturity model used here, current maturity is Level 2, Developing (A15). Basic plans exist: a warm-site drill with timings and offsite backups. However, there is no evidence of regular testing, an identity is shared between production and backup, provider controls are undocumented and two RPO gaps remain, so Level 3 is not yet reached. The realistic 12-month target is Level 3, Defined: an approved plan, quarterly testing, KPIs reported to the Steering Committee and at least 90% of corrective actions closed. Level 4 is not realistic within a year, because it needs a measured performance history and a third technician.

Table 26. Maturity assessment and target

Item	Current (Level 2 Developing)	12-month target (Level 3 Defined)
Plans	Warm-site drill and backups exist	Approved plan covering F1-F5 and controlled distribution
Testing	No evidence of regular testing	Calendar in Table 23 delivered; K3 at 100%
Governance	Shared identity; undocumented provider controls	CA1-CA4 closed; K4 at least 90%
Measurement	Not reported	Table 25 reported quarterly

References

[1] ICDFA, "Business Continuity and Disaster Recovery Assignment," GRC205 Incident Management and Business Continuity, Week 21, assignment brief. International Cybersecurity and Digital Forensics Academy.

[2] ICDFA, "Business Continuity and Disaster Recovery," GRC205 Incident Management and Business Continuity, Week 21, teaching slides 1-15. International Cybersecurity and Digital Forensics Academy.

[3] M. Swanson, P. Bowen, A. Phillips, D. Gallup, and D. Lynes, "Contingency Planning Guide for Federal Information Systems," NIST Special Publication 800-34 Rev. 1, National Institute of Standards and Technology, Gaithersburg, MD, USA, May 2010 (updated Nov. 11, 2010), doi: 10.6028/NIST.SP.800-34r1. [Online]. Available: https://csrc.nist.gov/pubs/sp/800/34/r1/upd1/final

[4] Federal Republic of Nigeria, “Nigeria Data Protection Act, 2023,” sec. 40. Nigeria Data Protection Commission, official hosted copy. [Online]. Available: https://ndpc.gov.ng/wp-content/uploads/2024/03/Nigeria_Data_Protection_Act_2023.pdf

[5] International Organization for Standardization, "ISO 22301:2019 Security and resilience - Business continuity management systems - Requirements," standard overview page. [Online]. Available: https://www.iso.org/standard/75106.html

[6] International Organization for Standardization, "ISO 22301:2019/Amd 1:2024 Security and resilience - Business continuity management systems - Requirements - Amendment 1: Climate action changes," Feb. 2024, standard page. [Online]. Available: https://www.iso.org/standard/88412.html

Appendix A Assumptions register

Table A1. Assumptions register (unverified scenario details)

ID	Assumption	Where used
A1	Northbridge operates in Nigeria (amounts in NGN, times in WAT), so the Nigeria Data Protection Act 2023 applies under s. 2(2)(a). The brief does not state the location.	4.5, 4.6
A2	All owner roles, names and contact addresses are fictional; .example addresses are used.	Tables 1, 11-17
A3	The exam length is not stated; it is assumed to be no more than 3 hours, so two sittings (09:00 and 14:00) fit within Day 1.	4.3, Tables 14, 20
A4	A same-day delayed start is not treated as an acceptable assessment service by 11:00; the NGN 120,000 trigger is incurred.	4.3
A5	Split sittings with approved paper variants are acceptable under Northbridge's assessment rules (the scenario prescribes no method).	4.3, Tables 14, 20
A6	The locations of the alternate facility, warm site and offsite provider relative to the flood area and power supply are not stated. They are assumed to be separate, and Facilities must confirm this before activation. The provider's access-independence remains unverified.	2.3, Table 8, 6.2
A7	Own-device learners can reach the restored LMS01 remotely through the recovery network.	Table 14
A8	Business acceptance takes 30 minutes per function after technical completion and is done by function owners in parallel, not by technicians.	Table 19, Appendix C
A9	B02 incrementals run hourly from 00:00 to 07:00.	Table 20, Appendix C
A10	The 17:00 Day -3 disconnected full backup contains DB01; whether it also holds AUTH01 and FS01 is unconfirmed.	Tables 5, 15, 20
A11	The staff listed in Table 13 are available during the outage, on site or remotely.	Table 13
A12	An incident response capability exists: an internal IR Lead plus an external retainer.	4.6, Table 20
A13	An SMS and voice call tree from staff mobile phones is available as the alternative channel.	Tables 11, 16
A14	Business staff can work on laptops through Microsoft 365 and remote access, leaving workstations for learners.	4.3
A15	No test history is provided; maturity is judged from the evidence in the supplied case information only.	6.3
A16	Illustrative only: hourly costs stop when the minimum service is declared.	3.3, Appendix C
A17	Calendar dates are fictional; target dates are counted from a plan approval date, P.	Table 21

Appendix C. Supporting checkpoint and scheduling calculations

Table C2. Checkpoint ages and inject 3 fallback (reference point 08:00 Day 0)

Item	Working	Age (h)	Gap against RPO
DB01 (B01 + B02 to 07:00)	08:00 - 07:00	1	1 - 1 = 0 (F1-F3)
AUTH01 (B03)	08:00 - 00:00	8	8 - 4 = 4 over
FS01 (B04)	1 h to midnight + 8 h	9	9 - 4 = 5 over
DB01 B01 only (inject 3, first incremental fails, A9)	1 h to midnight + 8 h	9	9 - 1 = 8 over
Disconnected full, 17:00 Day -3	7 h + 48 h + 8 h	63	63 - 1 = 62 over
Proposed DB01 chain (Table 6)	Sunday 23:00 full + 17 incrementals 23:30-07:30; 08:00 - 07:30	0.5	1 - 0.5 = 0.5 margin

Table C4. Schedule verification

Check	Result
Earliest DB01 finish	30 + 30 + 30 + 90 = 180 min, so 11:00
Lower bound for LMS01 finish	180 + 45 = 225 min, so 11:45
Lower bound for all functions	180 + (45 + 30 + 30 + 45) / 2 = 255 min, so 12:15
Analytical schedule verification	DB01 earliest 11:00; 150 technician-minutes for four application restores, divided across two technicians as 45+30 each. The schedule meets the 12:15 lower bound without overlap or interrupted actions.
Table 18 schedule	All prerequisites finish before each start; no technician overlap; no split actions; reaches both lower bounds
Exam gap	LMS01 finishes 45 min after the 11:00 exam start